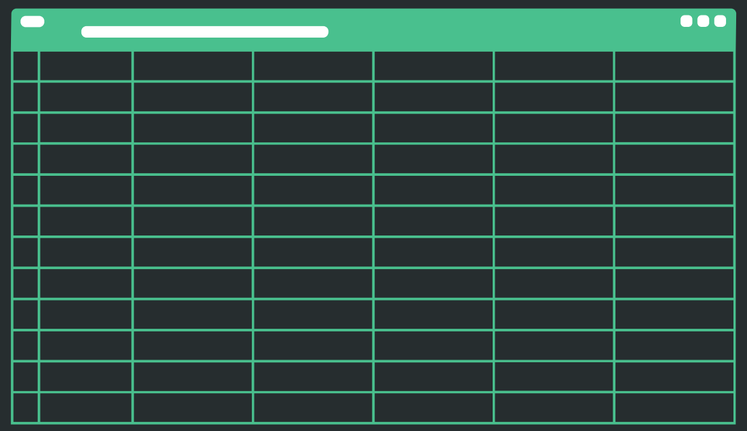
6.3. Linear Algebra - Basics

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Why Linear Algebra matters in ML

Foundation of Data

- Data in ML is numbers.
- Numbers live in vectors and matrices.
- Linear algebra lets models read and learn from them.



Numbers in ML: From Scalar to Tensor

Scalar

50

Single Number

Shape: ()

Vector

$$\begin{bmatrix} 5 \\ 23 \end{bmatrix}$$

1D Data

Shape: (n,)

Matrix

$$\begin{bmatrix} 10 & 33 \\ 5 & 1 \end{bmatrix}$$

2D Data

Shape: (m,n)

Tensor

$$\begin{bmatrix} 10 & 33 \\ 5 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$$

Multi - Dimensional

Shape: (d1, d2, d3, ...)

Why Shape Matters in ML

Machine Learning algorithms are essentially large math formulas.

For the math to work, the "plugs" must fit the "sockets". If dimensions don't align, the computation fails.

$$y = wX + b$$

Input matrix → X shape: (n_samples, n_features)

Weight vector → w shape: (n_features, 1)

Bias scalar → b shape: (1,)

Output prediction y → wX + b: (n_samples, 1)

Machine Learning - Dataset

Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	Diabetes Pedigree Function	Age	Outcome
1	85	66	29	0	26.6	0.351	31	0
2	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
1	137	40	35	168	43.1	2.288	33	1
3	116	74	0	0	25.6	0.201	30	0

The Big Picture: Datasets are matrices, samples & weights are vectors, and models are math operations.